A Comparison of Selected Organic Materials for Low Orbiting Spacecraft Hull Plating

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Abstract: Spacecraft operating in low Earth orbit (LEO) and during the launch phase are exposed to simultaneous extreme threats: atomic oxygen (AO) erosion, ultraviolet radiation, thermal cycling between $-150\,^{\circ}\mathrm{C}$ and $+150\,^{\circ}\mathrm{C}$, micrometeoroid and orbital debris (MMOD) impacts, and heat fluxes of 1–10 MW/m^2 during ascent. The selection of hull plating and thermal protection system (TPS) materials requires balancing low areal density, high structural strength, thermal stability, and surface durability. This paper presents a systematic comparison of thirteen organic materials (M1–M13) covering polyimide films, thermoset resins, elastomers, aramid fibres, and advanced composites. Mechanical and thermal properties at room temperature and as a function of temperature were compiled from manufacturer datasheets and peer-reviewed literature. Comparison charts and a five-criterion scoring methodology identify the top five candidate materials: Kapton HN (M1) and POSS-polyimide (M2) for LEO surface films; Kevlar-29/49 (M7) for structural and Whipple-shield applications; carbon-fibre/phenolic composite (M12) for high-heat-load TPS; and polysiloxane composite (M6) as a versatile ablative-structural hybrid.

Keywords: ablative materials, atomic oxygen, aramid fibres, Kapton, Kevlar, low Earth orbit, organic composites, polyimide, spacecraft, thermal protection system

Introduction

The protective plating of a spacecraft must satisfy radically different requirements during two successive mission phases. During launch and ascent, the vehicle is exposed to aerodynamic heat fluxes reaching 1–10 MW/m^2, acoustic loads up to 160 dB, and structural accelerations of 5–12 g (Natali et al. 2012, p. 1). Once in low Earth orbit at altitudes of 160–2000 km, the dominant threats change to atomic oxygen at fluences of 10^{20}–10^{21} atoms/cm^2 per year, vacuum-ultraviolet radiation, and thermal cycling between $-150\,^{\circ}\mathrm{C}$ and $+150\,^{\circ}\mathrm{C}$ with a period of approximately 90 minutes (Gouzman et al. 2010, p. 2438).

Organic materials occupy a strategic position in spacecraft design because they can be tailored through polymer chemistry and fibre reinforcement to address several of these requirements simultaneously at low areal density. Polyimide films provide radiation resistance, dimensional stability, and a wide service temperature range, making them the standard material for multi-layer insulation (MLI) blankets. Aramid fibres deliver specific tensile strengths exceeding 2.0 km^2/s^2, enabling lightweight Whipple shields and pressure vessels. Ablative composites pyrolyse controllably, consuming heat through endothermic reactions and leaving a protective char layer.

This paper compiles and analyses the key mechanical and thermal properties of thirteen organic materials selected on the basis of current or prospective use in spacecraft hulls and TPS. Materials are labelled M1–M13. The objective is to provide a single reference document that facilitates rational material selection for specific subsystems, supported by temperature-dependent data covering the full LEO thermal envelope.

Environmental Conditions

The design envelope for spacecraft hull materials is defined by two overlapping phases summarised in Table 1.

Table 1. Design conditions for spacecraft hull materials during launch and LEO service.

Phase	Primary threats	Key material requirements
Launch / ascent	Heat flux 1–10 MW/m^2; vibration; acceleration 5–12 g	Ablation / TPS; high mechanical strength; low areal density
LEO (orbit)	AO 10^{20}–10^{21} at/cm^2/yr; UV; thermal cycling -150 to $+150\,^{\circ}$C; MMOD	AO/UV resistance; dimensional stability; $K_{\mathrm{IC}} > 1\,\mathrm{MPa\,m^{0.5}}$

Atomic oxygen is the dominant chemical threat below 600 km altitude. It erodes most organic polymers at rates of 10^{-24}–10^{-23} cm^3/atom. Kapton erodes at approximately 3×10^{-24} cm^3/atom; aliphatic polymers such as UHMWPE erode 3–5 times faster due to the absence of aromatic rings in the backbone. POSS-modified polyimides and polysiloxane composites benefit from passive self-passivation: AO oxidises surface silicon atoms to SiO_2, forming a self-healing ceramic barrier (Gouzman et al. 2010, p. 2440).

Thermal cycling introduces cyclic mechanical strains proportional to the coefficient of thermal expansion (CTE) mismatch between bonded layers. For a polymer film adhered to a metallic structure, strains per cycle can reach 0.1–0.5%, requiring high fatigue resistance and fracture toughness.

Materials

Polyimide Films

Kapton HN (M1) is a poly(4,4′-oxydiphenylene-pyromellitimide) film produced by DuPont and used on spacecraft since the Apollo programme. Its room-temperature tensile strength (UTS) of 231 MPa at a density of 1420 kg/m^3 gives a specific strength of 163 kN·m/kg. Young's modulus is 2.5 GPa and fracture toughness is 1.65–5.4 MPa m$^{0.5}$

depending on film thickness and orientation (DuPont 2023). The continuous service range of -269 to $+400\,^{\circ}\mathrm{C}$ covers both cryogenic propellant tanks and aerodynamic heating. Thermal conductivity is 0.12 W/(m·K). The principal LEO limitation is AO erosion, which requires a protective SiO_2 or Al coating on sun-facing surfaces.

POSS-Polyimide (M2) incorporates polyhedral oligomeric silsesquioxane (POSS) nanocages into the polyimide backbone. AO oxidises the surface Si atoms to SiO_2, forming a self-passivating ceramic layer that suppresses further erosion — the key advantage over Kapton for long-duration LEO missions. UTS is approximately 210 MPa, $E = 2.3$ GPa, and operating range is -269 to $+450\,^{\circ}\mathrm{C}$ (Brunsvold et al. 2004, p. 303; Gouzman et al. 2010, p. 2439).

Thermoset Resins

Phenol-Formaldehyde Resin (M3a) — phenol-formaldehyde thermosetting resins have been the matrix material of choice for ablative TPS since the 1960s. Neat resin has UTS = 45 MPa, E = 3.5 GPa, and a brittle failure mode (elongation 1%). At temperatures above 300 °C the resin pyrolyses, forming a porous char with approximately 55% carbon yield, which insulates the underlying structure. The maximum effective ablative protection temperature exceeds 2000 °C (Natali et al. 2012, p. 3). Density is $1250\,\mathrm{kg/m^3}$.

Phthalonitrile Resin (M4) represents the next generation of high-temperature thermosets. Cure proceeds without volatile byproducts, yielding void-free, low-outgassing parts — a critical requirement near spacecraft optics and sensors. The glass transition temperature T_g exceeds 375 °C, the highest continuous service temperature of any unreinforced polymer in this study. UTS = 65 MPa, E = 4.0 GPa; char yield is approximately 65% (Hergenrother et al. 2005, p. 7853; Laskoski et al. 2006, p. 4559).

Elastomers and Composite Films

RTV Silicone (M5) — room-temperature-vulcanising polydimethylsiloxane elastomers (e.g. Momentive RTV566) are used as structural adhesives, sealants, and vibration dampers in satellite assemblies. UTS is only 6 MPa but elongation reaches 400%, giving a uniquely hyperelastic stress–strain response. The service range is -115 to $+300\,^{\circ}\mathrm{C}$; at cryogenic temperatures Young's modulus rises sharply from 3 MPa at room temperature to approximately 2000 MPa at $-115\,^{\circ}\mathrm{C}$ (Barucci et al. 1998, p. 3).

SiO_2f/SiO_2 Polysiloxane Composite (M6) — 2.5D SiO_2-fibre/SiO_2-matrix composites (McDermott et al. 2022) combine ablative performance with structural integrity. A char yield of 86.5% and the formation of an SiOC ceramic matrix maintain structural function at temperatures above 1400 °C. Room-temperature properties: UTS = 182 MPa, E = 45.5 GPa, density = $1320\,\mathrm{kg/m^3}$, K_IC = 2.52 MPa $\mathrm{m^{0.5}}$. AO resistance is inherently high due to the silica-rich surface.

Aramid Fibres

Kevlar (M7) — DuPont's para-aramid fibres are produced in two principal grades. *Kevlar-29* (M7a): UTS = 3600 MPa, E = 70.5 GPa, elongation 3.6%. *Kevlar-49* (M7b): UTS = 3800 MPa, E = 125 GPa, elongation 2.4%. Both grades have density 1440 kg/m^3 and retain more than 75% of room-temperature UTS across the range -196 to $+430\,^\circ$C (DuPont 2017). Thermal conductivity is exceptionally low at 0.04 W/(m·K), dropping to 0.007 W/(m·K) at 7 K (Ventura & Martelli 2009, p. 735). Kevlar-29 is preferred for Whipple shields and pressure vessels where energy absorption governs; Kevlar-49 for stiffness-critical structures such as optical benches.

PET Film

Mylar BoPET (M8) — biaxially oriented poly(ethylene terephthalate) film from DuPont Teijin Films is the standard reflective layer in MLI blankets, typically aluminised on one or both sides. UTS = 200 MPa, E = 3.95 GPa, density = 1395 kg/m^3, K_{IC} = 3.5 MPa m$^{0.5}$. The glass transition at $T_{\mathrm{g}} \approx 80\,^\circ$C causes UTS to fall sharply above this temperature, restricting structural service to the range -70 to $+150\,^\circ$C (DuPont Teijin Films 2021).

Ultra-High-Molecular-Weight Polyethylene and Composites

M9 and M10 represent two forms of the same base polymer — ultra-high-molecular-weight polyethylene (UHMWPE, molecular weight 3.5–7.5 $\times$ 10^6 g/mol) — in bulk and fibre-reinforced composite form respectively.

UHMWPE bulk (M9) — solid UHMWPE has a density of only 940 kg/m^3, making it the lightest material in this set and the only one less dense than water. Room-temperature ballistic performance rivals Kevlar. The critical limitation is a melt temperature of 135 °C: UTS falls to approximately 50 MPa already at 80 °C, ruling M9 out of any thermally exposed structural role. The purely aliphatic backbone offers no AO resistance (Kurtz 2009, p. 23; Sobieraj & Rimnac 2009, p. 433).

UHMWPE Fibre Laminate (M10) — gel-spun UHMWPE fibres (Dyneema SK75/SK76, Spectra 1000) achieve a fibre-axis modulus of 70–150 GPa through near-perfect chain alignment. Consolidated as cross-plied laminates in an epoxy or HDPE matrix they deliver UTS = 400 MPa and E = 30 GPa at a density of 1000 kg/m^3 — roughly double the strength and 33 times the stiffness of bulk M9. The trade-off is identical thermal and AO sensitivity: service temperature is limited to 120 °C by the matrix, and the aliphatic fibre surface remains susceptible to atomic oxygen (Koh et al. 2010, p. 2234).

Advanced Structural Composites

Kevlar/Epoxy Composite (M11) — woven Kevlar-49 fabric in Hexcel 8552 epoxy resin (fibre volume fraction $\approx$ 60%) achieves UTS = 600 MPa, E = 40 GPa, density

= 1380 kg/m^3. UTS is thermally stable from -55 to $+180$ °C. Used for pressure vessels, primary load-bearing panels, and satellite structural frames (Duan et al. 2006, p. 441).

Carbon-Fibre/Phenolic Composite (M12) — carbon-fibre-reinforced phenolic composites constitute the primary ablative TPS for solid-rocket-booster (SRB) nozzles and high-heat-load atmospheric re-entry vehicles. Carbon-fibre reinforcement raises UTS to 350 MPa and E to 35 GPa compared to neat phenolic resin. The char layer above 2000 °C is extremely stable. Thermal conductivity $k = 2.0\,\mathrm{W/(m \cdot K)}$ — the highest of all thirteen materials — facilitates heat redistribution in thick TPS stacks. K_{IC} = 15 MPa m$^{0.5}$ is also the highest in the set (Natali et al. 2012, p. 5; Tran et al. 2014, p. 1418).

CF/Polysiloxane Composite (M13) — carbon-fibre-reinforced polysiloxane-matrix composites with lower reinforcement fraction achieve UTS = 150 MPa, E = 25 GPa, density = 1450 kg/m^3, and effective ablative protection to 1200 °C (McDermott et al. 2022, p. 9).

Results and Discussion

Summary of Room-Temperature Properties

A summary of all key room-temperature properties for M1–M13 is given in Figure 1. The data span four orders of magnitude in UTS (6–3800 MPa), six orders of magnitude in Young's modulus (0.003–125 GPa), and more than two orders of magnitude in specific strength.

Tensile Strength

The UTS comparison in Figure 2 is plotted on a logarithmic scale because the data span four decades. Kevlar fibres (M7a, M7b) are dominant at 3600–3800 MPa — more than six times the UTS of the next structural composite (Kevlar/epoxy M11, 600 MPa). At the lower end, the neat thermoset resins M3 and M4 record 45–65 MPa, and the RTV elastomer M5 only 6 MPa.

Young's Modulus and Stiffness

The stiffness hierarchy (Figure 3) follows a similar pattern. Kevlar-49 (M7b, 125 GPa) is the stiffest organic material known, approaching the modulus of aluminium alloys at one-third the density. SiO_2f/SiO_2 polysiloxane composite (M6, 45.5 GPa) and Kevlar/epoxy (M11, 40 GPa) represent practical laminate options when stiffness-to-weight ratio governs the design. RTV silicone (M5, 0.003 GPa) is six orders of magnitude more compliant — its role is vibration damping and sealing, not structural load bearing.

Material	UTS [MPa]	E [GPa]	ρ [kg/m³]	T_min [°C]	T_max [°C]	k [W/m·K]	K_IC [MPa√m]
M1 Kapton	231	2.50	1 420	**−269**	400	0.12	3.5
M2 POSS-Polyimide	210	2.30	1 450	**−269**	450	0.15	2.0
M3 Ph-F Resin (M3a)	45	3.50	1 250	−55	2000*	0.30	0.7
M4 Phthalonitrile	65	4.00	1 250	−55	375	0.20	1.0
M5 RTV Silicone	6	0.003	1 175	−115	300	0.25	—
M6 SiO2f/SiO2 Compos.	182	45.5	1 320	−60	1400*	0.21	2.52
M7a Kevlar-29	**3 600**	70.5	1 440	**−196**	430	0.04	—
M7b Kevlar-49	**3 800**	125	1 440	**−196**	430	0.04	—
M8 Mylar BoPET	200	3.95	1 395	−70	150	0.15	3.5
M9 UHMWPE	200	0.90	940	−150	80	0.44	2.0
M10 UHMWPE Lam.	400	30.0	1 000	−150	120	0.35	—
M11 Kevlar/Epoxy Comp.	600	40.0	1 380	−55	180	0.12	—
M12 CF/Phenolic Compos. (M12)	350	35.0	1 550	−55	2000*	2.00	**15**
M13 CF/Polysilox. Compos.	150	25.0	1 450	−60	1200*	0.50	2.0

* T_max for ablative materials = effective ablative protection, not continuous structural service.

Figure 1. Room-temperature material properties for M1–M13. Green cells denote top-performance values: UTS $\geq$ 3000 MPa, $T_{\min} \leq -196$ °C, $K_{\mathrm{IC}} \geq 10$ MPa$\sqrt{\mathrm{m}}$. Asterisk denotes effective ablative service temperature, not continuous structural service.

Density and Specific Strength

Figure 4 shows densities ranging from 940 kg/m^3 (UHMWPE, M9) to 1550 kg/m^3 (carbon/phenolic composite, M12). The narrow density range is a characteristic of organic materials and contrasts sharply with metals (2700–8900 kg/m^3).

Specific strength (Figure 5) reveals the true competitive advantage of Kevlar: at 2.5 km^2/s^2, it exceeds carbon-fibre/phenolic composite (M12) by a factor of seven and polyimide film (M1) by a factor of fifteen. For mass-critical structural applications, Kevlar fibres have no organic competitor.

Service Temperature Range

Figure 6 presents service temperature ranges. Ablative materials (M3, M6, M12, M13) cover the widest thermal envelopes, with effective protection extending above 1000 °C via char or SiOC-ceramic formation. Kapton (M1) and POSS-PI (M2) cover the widest continuous service range (-269 to $+400/450$ °C), encompassing both the cryogenic propellant temperature and the aerodynamic heating regime. The LEO thermal cycling envelope of -150 to $+150$ °C is covered by all materials except UHMWPE (limited to $+80$ °C), which is therefore unsuitable for external LEO surfaces.

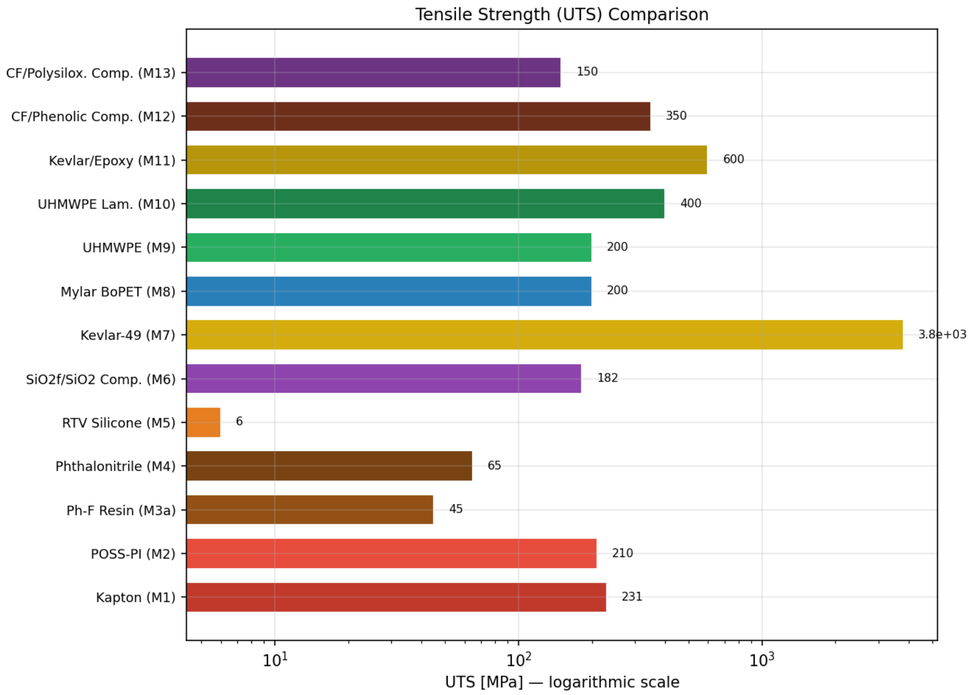

Figure 2. Tensile strength (UTS) comparison of M1–M13, logarithmic scale.

Material Selection for Spacecraft Applications

Scoring Methodology

Five criteria are used to score each material for combined launch-phase and LEO service. Weights reflect the relative severity of each threat in a typical LEO mission:

- Specific strength UTS/ρ 25%
- Service temperature range $T_{max} - T_{min}$ 20%
- Resistance to atomic oxygen and UV radiation 20%
- Heat absorption / ablative capacity 20%
- Fracture toughness K_{IC} / impact resistance 15%

Each criterion is normalised to the interval $[0, 1]$ across all thirteen materials and multiplied by the criterion weight. The resulting scores are visualised in the radar chart (Figure 7).

Top-5 Recommended Materials

1. Kapton HN (M1) — Standard LEO Surface Film

Kapton's 60-year space heritage, zero creep in vacuum, and service range of −269 to +400 °C make it the default choice for MLI blankets, flexible printed circuits,

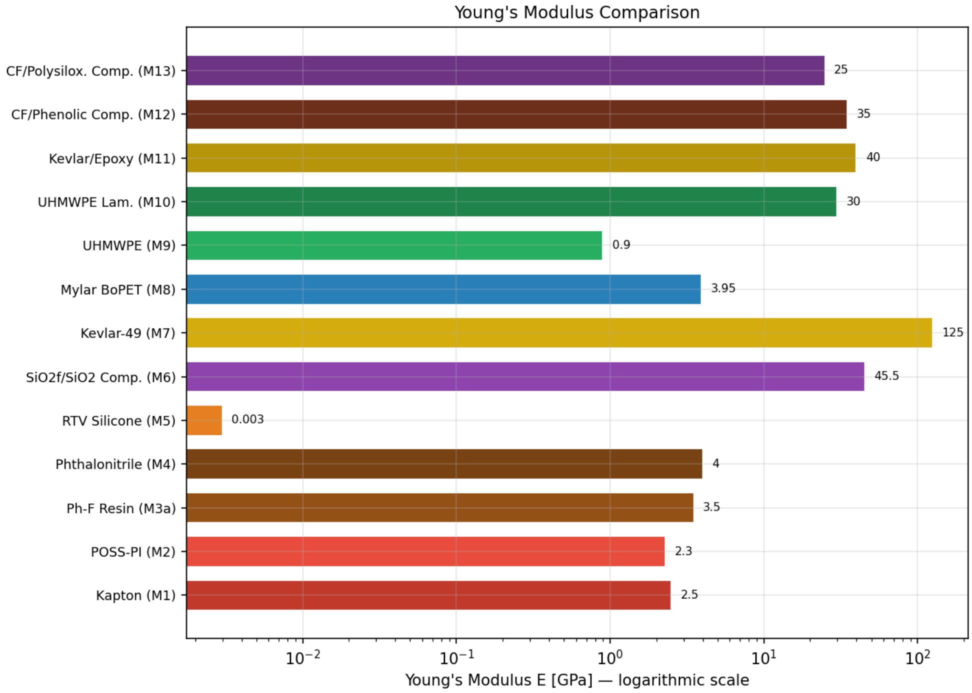

Figure 3. Young's modulus comparison of M1–M13, logarithmic scale.

and thermal control surfaces. It is the only material in this set with flight-validated performance on every class of spacecraft from CubeSats to planetary probes. The AO erosion limitation is well-understood and addressed by thin-film coatings (DuPont 2023).

2. Kevlar-29/49 (M7) — Structural and Impact-Protection Fibre

Specific strength of 2.5 km^2/s^2 and high energy-absorption capacity during hyper-velocity impact (via fibre pull-out and fracture) make Kevlar the baseline material for Whipple shields, pressure vessels, and load-bearing panels. Temperature stability across the full LEO envelope (-196 to $+430$ °C) and decades of in-service data support its selection (DuPont 2017).

3. SiO_2f/SiO_2 Polysiloxane Composite (M6) — Ablative-Structural Hybrid

M6 is the most versatile material in the set: UTS = 182 MPa and E = 45.5 GPa provide genuine structural load-bearing capacity alongside ablative TPS protection to 1400 °C. The SiO_2-rich surface passivates against AO without an external coating. This combination makes it suited for nozzle liners, re-entry nose-cap segments, and interstage thermal protection structures.

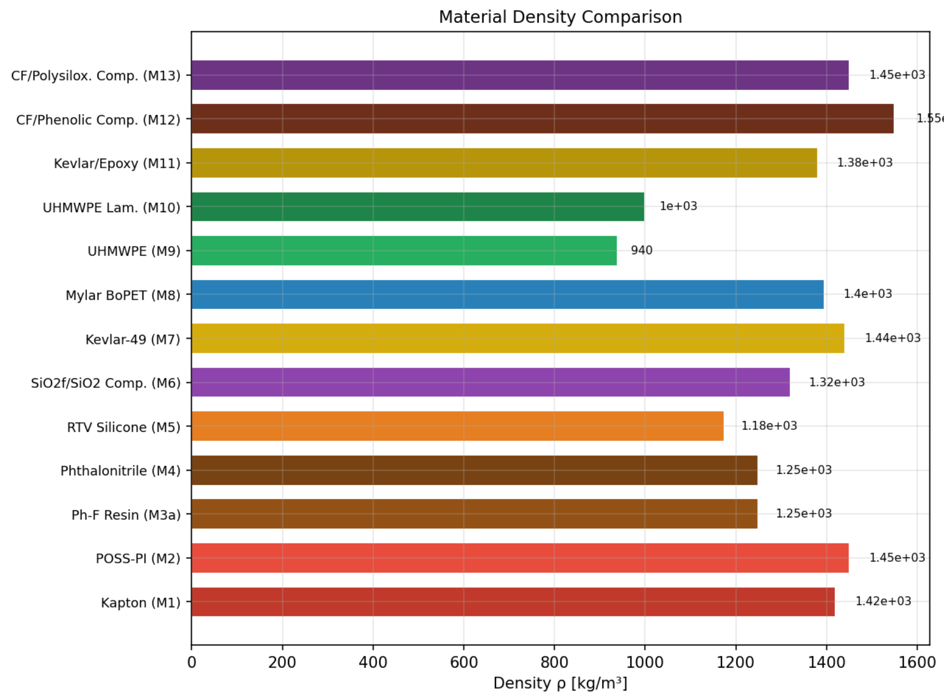

Figure 4. Density comparison of M1–M13. UHMWPE (M9) is the only material with density below that of water.

4. Carbon-Fibre/Phenolic Composite (M12) — Primary High-Heat-Load TPS

For applications where heat flux exceeds 5 MW/m^2 (SRB nozzles, re-entry heat shields), the carbon-fibre/phenolic composite has no organic competitor. Its K_{IC} = 15 MPa m$^{0.5}$ prevents catastrophic crack propagation under thermal shock, and k = 2.0 W/(m·K) distributes heat laterally in thick TPS assemblies (Tran et al. 2014, p. 1418).

5. POSS-Polyimide (M2) — Next-Generation LEO Surface Film

POSS-PI surpasses Kapton in AO resistance and upper service temperature ($+450\,°$C vs. $+400\,°$C), making it the preferred choice for long-duration ($>$ 5-year) LEO missions and any surface where SiO_2 coating deposition is impractical. Currently higher in cost than Kapton, it is deployed selectively on highest-priority external surfaces (Brunsvold et al. 2004, p. 303).

Conclusions

A systematic comparison of thirteen organic materials for LEO spacecraft hull plating and thermal protection has been presented. The principal findings are:

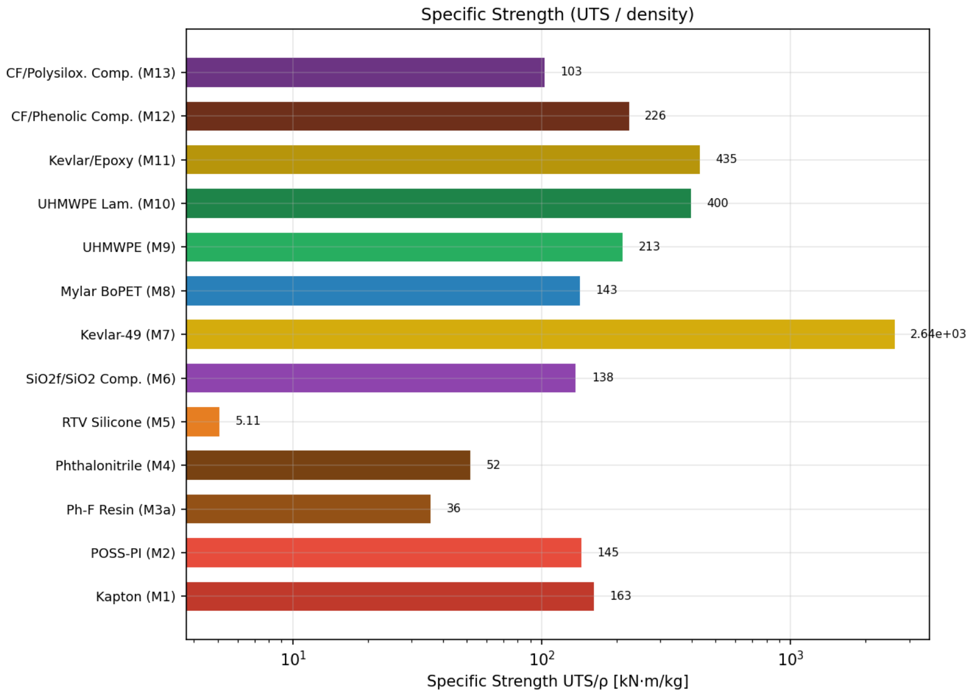

Figure 5. Specific strength (UTS/ρ) comparison of M1–M13.

1. Kevlar fibres (M7a, M7b) are the only materials in the set with specific strength exceeding 2.0 km^2/s^2, making them irreplaceable for mass-critical structural and impact-protection applications.
2. Polyimide films (M1, M2) provide the widest continuous service temperature range and the best combination of AO/UV resistance and zero vacuum outgassing, qualifying them as the standard LEO surface material.
3. Ablative composites (M3, M6, M12, M13) are essential for launch and re-entry phases where heat fluxes exceed 1 MW/m^2; carbon-fibre/phenolic (M12) offers the highest protection capacity and fracture toughness.
4. UHMWPE-based materials (M9, M10) offer attractive ballistic performance at low density but are disqualified from external LEO surfaces by their low melt temperature (≤ 135 °C) and AO susceptibility.
5. No single material satisfies all requirements simultaneously. Practical spacecraft designs combine two or more materials from this set — for example, a Kapton MLI blanket over a Kevlar Whipple shield backed by a phenolic ablator — to achieve multi-functional performance across both mission phases.

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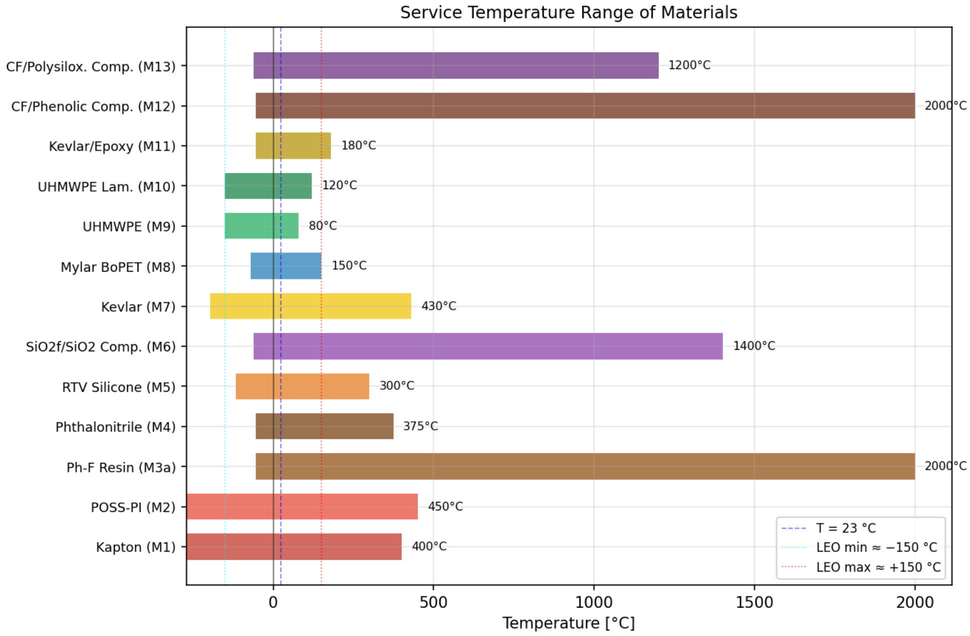

Figure 6. Service temperature range of M1–M13. Dotted lines mark the LEO thermal envelope (-150 to $+150$ °C).

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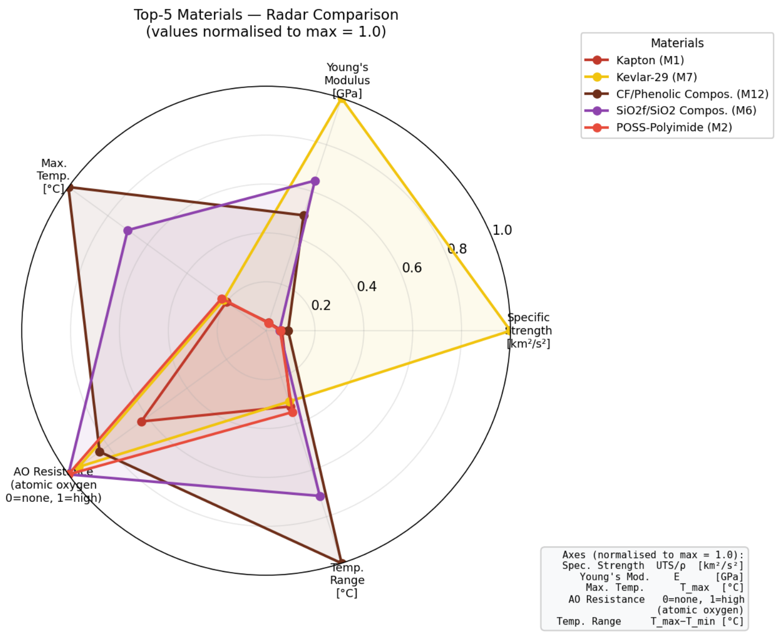

Figure 7. Multi-criteria radar comparison of the Top-5 candidate materials (normalised values, $\max = 1.0$***).***

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Porównanie wybranych materiałów organicznych stosowanych jako poszycie kadłuba statku kosmicznego na niskiej orbicie okołoziemskiej

Streszczenie: Statki kosmiczne operujące na niskiej orbicie okołoziemskiej (LEO) oraz podczas startu narażone są na jednoczesne ekstremalne zagrożenia: erozję wywołaną atomowym tlenem (AO), promieniowanie ultrafioletowe, cykle termiczne w zakresie od $-150\,^{\circ}$C do $+150\,^{\circ}$C, uderzenia mikrometeorytów i szczątków orbitalnych, a podczas wznoszenia — strumienie ciepła 1–10 $\mathrm{MW/m^2}$. Dobór materiałów poszycia i systemów ochrony termicznej wymaga kompromisu między niską gęstością powierzchniową, wytrzymałością mechaniczną, stabilnością termiczną i trwałością powierzchniową. W artykule przedstawiono systematyczne porównanie trzynastu materiałów organicznych (M1–M13) obejmujących folie poliimidowe, żywice termoutwardzalne, elastomery, włókna aramidowe oraz zaawansowane kompozyty. Właściwości mechaniczne i termiczne zestawiono na podstawie kart katalogowych producentów i recenzowanych publikacji naukowych. Wielokryterialna metoda oceny pozwoliła wyłonić pięć najlepszych materiałów: Kapton HN (M1) i POSS-poliimid (M2) jako folie powierzchniowe LEO, Kevlar-29/49 (M7) do zastosowań strukturalnych i tarcz Whipple'a, kompozyt węglowo-fenolowy (M12) jako TPS wysokiego strumienia ciepła oraz kompozyt polisiloksanowy (M6) jako wielofunkcyjny materiał ablacyjno-strukturalny.

Słowa kluczowe: kompozyty organiczne, Kapton, Kevlar, materiały ablacyjne, niska orbita okołoziemska, poliimid, statek kosmiczny, system ochrony termicznej, tlen atomowy, włókna aramidowe